

If an Ubuntu Linux laptop has an HDMI connected monitor connected to it, then the i3 window manager places a virtual desktop, for example, virtual desktop number 6, either on laptop screen or the monitor. How to make it so that a single virtual desktop always spans both, laptop screen and the attached monitor?

Short answer: you can't make a single i3 "workspace" truly *span* two independent monitors. That's not how i3 window manager is designed.

In i3, each workspace exists on exactly **one output at a time** (one screen). You can move it between outputs, but it won't stretch across both simultaneously.

What you *can* do instead

1. Treat both screens as one big display (the only real "span" option)

You can merge your laptop screen + HDMI monitor into a single large virtual screen using xrandr:

```
</> Bash
xrandr --output HDMI-1 --right-of eDP-1
```

Then create a framebuffer that covers both:

```
</> Bash
xrandr --fb 3840x1080
```

(Adjust resolution to match your combined screens.)

👉 Result:

- i3 sees **one big screen**
- Workspaces now "span" both physically

⚠️ Downsides:

- Windows can cross the seam awkwardly
- Fullscreen apps will cover both screens

- Different resolutions/scaling can get messy
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2. Assign matching workspaces per screen (most common approach)

Instead of spanning, people simulate it:

```
</> Bash
workspace 6 output HDMI-1
workspace 6 output eDP-1
```



Then use scripts or keybindings to switch both outputs together.

You can also bind something like:

```
</> Bash
bindsym $mod+6 workspace 6; move workspace to output right
```



But note: it still moves, not duplicates.

3. Use two synchronized workspaces (scripted workaround)

Advanced users create paired workspaces like:

- 6-left
- 6-right

And switch both at once via script (using i3 IPC via `i3-msg`).

This gives the *feeling* of one workspace across two screens without breaking i3's model.

4. Consider a different WM if true spanning is critical

Some environments treat multi-monitor setups differently. For example:

- GNOME (with extensions)
- KDE Plasma
- awesome (more customizable behavior)

But even most tiling WMs follow the “workspace per output” model.

Bottom line

- i3 does **not support a workspace spanning multiple outputs**
 - The only real workaround is faking a single display via xrandr
 - Otherwise, you simulate it with coordinated workspaces
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If you want, tell me your exact monitor resolutions and layout—I can give you a precise `xrandr` setup or a clean i3 config snippet tailored to your setup.

